

Hybrid Scheduling Heuristic (ILP + Local Rules)

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1 Introduction

This work presents a constructive heuristic for the employee scheduling problem with minimum hourly requirements per team. The heuristic was designed as an alternative to exact methods (ILP and CSP), which proved to be computationally expensive for large temporal instances.

The main idea consists of decomposing the problem into two levels:

1. Exact determination of the minimum number of work blocks required per team and per day;
2. Heuristic assignment of these blocks to employees, respecting labor rules and promoting workload balance.

The heuristic constructs the solution sequentially over the planning horizon, ensuring feasibility in all intermediate decisions.

2 General Steps of the Heuristic

The heuristic follows the main steps below:

1. Initialization of the calendar, employees, teams, and work blocks;
2. For each day in the planning horizon:
 - (a) Verification of closed days (holidays and Sundays);
 - (b) Exact computation of the minimum set of required blocks per team;
 - (c) Creation of a global employee assignment ordering;
 - (d) Iterative assignment of blocks to eligible employees;
 - (e) Update of counters and scores;
3. Export of the final solution.

3 Indices

Set	Description
T	Set of available teams
H	Set of store operating hours
B	Set of valid work blocks
B_{dt}	Set of work blocks to be assigned in that day d for that team t
T_e	Set of teams, in which employee e can work

4 Parameters

Parameter	Description
θ_{dht}	Minimum number of employees of team t at hour h on day d
δ_{ed}	1 if employee e is on vacation on day d , 0 otherwise

5 State Variables

Variable	Description
W_e	Total number of days worked by employee e
W_e^w	Number of days worked by employee e sequentially
P_e	Current score of employee e
L_e	Block worked by employee e on the previous day

6 Rules and Constraints Covered

The heuristic guarantees compliance with the following rules:

- Full coverage of hourly minimum requirements per team (Soft Constraint);
- At most one work block per employee per day;
- At most 5 working days per week;
- At most 223 working days per year;
- Respect for vacations, holidays, and Sundays;
- Minimum rest of 12 hours between consecutive working days;
- Assignments compatible with employee team qualifications only.

7 Auxiliary Functions of the Heuristic

7.1 Generation of Work Blocks

Work blocks represent 8-hour shifts with a 1-hour break, covering two continuous working periods. Each block is defined by a triple $(h_{start}, h_{break}, h_{end})$.

7.2 Exact Computation of Required Blocks

For each day and team, a *multi-cover* problem is solved via **ILP**, whose objective is to minimize the number of blocks required to cover hourly minimum demands. See Appendix A for the formulation.

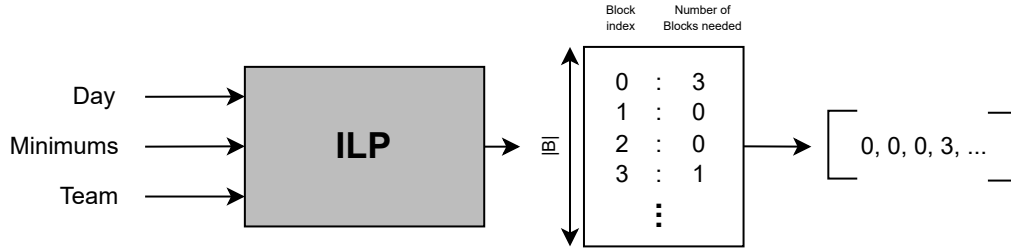


Figure 1: Computation of Blocks Process

7.3 Day to Blocks for Each Team

This function computes the ILP function for each team, and orders the blocks, from the earliest to the latest.

The result is the following:

$$\left\{ \begin{array}{l} \text{"A"} : [0, 0, 0, 3, \dots], \\ \text{"B"} : [0, 1, 1, 5, \dots], \\ \dots \end{array} \right\}$$

Figure 2: Final Structure for Blocks Distribution

7.4 Employee Scoring Function

The scoring function defines the assignment priority of each employee.

$$P_e = w_1 \left(1 - \frac{W_e}{223} \right) + w_2 \left(1 - \frac{W_e^w}{5} + 0.2\delta_{ed} \right)$$

where:

- W_e represents the total number of days worked;
- W_e^w represents the number of days worked in the current week;
- δ_{ed} represents a binary variable that checks whether employee e is on vacations on day d
- w_1 and w_2 are tunable weights.

7.5 Global Assignment Order

The global assignment order is determined by sorting all employees according to their score in P , in descending order. In the event of a tie (employees with the same score), the order among them is decided randomly at each execution.

This approach ensures a fair and impartial selection, giving equal opportunity to employees with identical scores, regardless of their team affiliations. This allows the algorithm to search for a wide range of solutions, that could be better or worse than a deterministic one.

The result is a single, unique, and globally ordered list of employees for assignment.

8 Heuristic Pseudocode

Algorithm 1 Schedule Assignment Heuristic

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1: for each day  $d$  in the planning horizon do
2:   if  $d$  is closed (Sunday or Holiday) then
3:     continue
4:   end if
5:   Compute hourly minimum requirements  $\theta_{d,h,t}$ 
6:   Compute Blocks for each Day and Team
7:   Build Global Employee Order from  $P$ 
8:   for each employee  $e$  in the global order do
9:     if  $e$  is not eligible then
10:      Reset  $L_e$  and  $W_e^w$ 
11:      continue
12:    end if
13:    if  $B_{dt} \in \emptyset$  then
14:      continue
15:    end if
16:    Compute the most needed teams in which Employee  $e$  can work
17:    for each compatible team in  $T_e$  do
18:      if Employee  $e$  has been assigned a block then
19:        Break
20:      end if
21:      for each available block in  $B_{dt}$  do
22:        if assignment is valid - 12 Hours Rule then
23:          Assign block
24:          Pop Block from  $B_{dt}$ 
25:          Update state variables excluding  $P_e$ 
26:          break
27:        end if
28:      end for
29:    end for
30:    if Not Assigned then
31:      Reset  $L_e$  and  $W_e^w$ 
32:    end if
33:    Compute Score  $P_e$  for employee  $e$ 
34:  end for
35: end for

```

9 General Structure of the Heuristic

The heuristic follows a sequential constructive strategy per day, always guaranteeing the feasibility of local decisions and full coverage of global requirements.

The use of a small-scale ILP solely to determine the minimum number of blocks enables a hybrid solution: exact in dimensioning and heuristic in allocation.

A ILP Formulation

A.1 Problem Description

The objective is to select a multiset of work blocks such that the demand for every hour is covered, while minimizing the total number of selected blocks.

Each block may be selected multiple times.

A.2 Sets and Indices

Symbol	Description
$b \in B$	Set of available work blocks
$h \in H$	Set of working hours
$H^+ \subseteq H$	Subset of hours with positive demand

A.3 Parameters

Parameter	Description
θ_h	Demand required at hour $h \in H^+$
α_{bh}	1 if block b covers hour h , 0 otherwise

A.4 Decision Variables

Variable	Description
$x_b \in \mathbb{Z}_0^+$	Number of times block $b \in B$ is selected

A.5 Objective Function

Minimize the total number of selected blocks

$$\min \sum_{b \in B} x_b \quad (1)$$

A.6 Constraints

Hourly Coverage Constraints

$$\sum_{b \in B} \alpha_{bh} x_b \geq d_h \quad \forall h \in H^+ \quad (2)$$

For each hour with positive demand, the total number of selected blocks covering that hour (i.e $\alpha_{bh} = 1$) must be at least the required demand (d_h)